

The Hawkes Paper: Markovian and Non-Markovian Self-Exciting Point Processes for Toronto Crime

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Abstract

Ruhela study likelihood-based inference for Hawkes self-exciting point processes applied to incident-level open data from the Toronto Police Service. The classical Mohler–Bertozzi–Brantingham model assumes a constant baseline ν and an exponential excitation kernel $g(t) = \beta e^{-\beta t}$; under those assumptions the bivariate process (N_t, λ_t) is Markovian, which is what enables the standard consistency and asymptotic-normality proofs. Building on Kwan, Chen, and Dunsmuir (2024) Ruhela drop both assumptions: the baseline $\nu(t)$ is allowed to vary in time, and the excitation kernel $g(\cdot)$ may be Gamma, Weibull, or power-law, all of which make the resulting intensity process *non-Markovian*. Ruhela give the full likelihood, score, and information for the general model, summarise the asymptotic-ergodicity argument that secures strong consistency and asymptotic normality of the MLE, and implement all of it as the new `moirais.tps_hawkes_advanced` module. Eight (kernel $\times$ baseline) combinations are fit to each TPS incident category and ranked by AIC and time-rescaling goodness-of-fit. For Toronto Assault data, the non-Markovian Gamma-kernel $\times$ sinusoidal-baseline model is preferred over the Markovian classical Hawkes by $\Delta\text{AIC} > 0$ and a higher Kolmogorov–Smirnov p -value on the residual uniforms.

1 Introduction

A Hawkes process [1] is a counting process $\{N_t\}_{t \geq 0}$ whose conditional intensity exhibits self-excitation: the occurrence of an event at $s < t$ raises $\lambda(t)$ above its baseline. For street-level crime, this self-excitation is the canonical mechanism of *near-repeat victimisation* — a burglary on a block raises the rate of further burglaries in the same block over the following days [4, 5]. Ruhela follow the spatial-physics-of crime literature in writing

$$\lambda_t = \nu + \int_{-\infty}^{t-} g(t-s) \, dN_s, \quad (1)$$

where ν is the baseline and g the excitation kernel. The classical Mohler model takes ν constant and $g(t) = \eta \beta e^{-\beta t}$; with the additional decomposition $g(t) = \eta \tilde{g}(t)$ where $\eta = \int_0^\infty g(t) \, dt$ is the *branching ratio* and $\tilde{g}$ is a probability density on $[0, \infty)$, stationarity requires $\eta < 1$ [3]. By the memorylessness of the exponential the bivariate (N_t, λ_t) is Markov [2], which is the key technical reason the classical likelihood inference is well understood.

The classical assumptions are convenient but restrictive. Trading intensity in financial markets is higher near the open and close [7, 14]; epidemic incidence has a latent period that the exponential kernel cannot represent [8, 9]; limit-order books exhibit power-law decay rather than exponential [10, 11]. In urban crime data Ruhela likewise expect (i) a time-varying baseline that follows seasonality and trend, and (ii) a kernel whose mode is shifted away from $t = 0$. The Kwan-Chen-Dunsmuir framework, built on [15] and infill asymptotics, makes likelihood inference for this general model rigorous. The contribution is to (a) re-derive the relevant finite-sample quantities cleanly, (b) implement them in `moirais.tps_hawkes_advanced`, and (c) apply the full machinery to Toronto Police Service open data.

2 Setup and notation

Let $0 \leq t_1 < t_2 < \dots < t_n \leq T$ be the observed event times of a simple point process N on $[0, T]$ with $\mathcal{F}_t = \sigma\{N_s : 0 \leq s \leq t\}$ -conditional intensity

$$\lambda(t) = \nu(t) + \int_0^{t-} g(t-s) dN_s = \nu(t) + \sum_{j:t_j < t} g(t-t_j). \quad (2)$$

The kernel admits the decomposition $g(u) = \eta \tilde{g}(u; \psi)$ with $\eta \in (0, 1)$ the branching ratio and $\tilde{g}(\cdot; \psi)$ a parametric density on $[0, \infty)$. Ruhela consider four kernel families:

$$\tilde{g}_{\text{exp}}(u; \beta) = \beta e^{-\beta u}, \quad \beta > 0, \quad (3)$$

$$\tilde{g}_{\text{gamma}}(u; \alpha, \beta) = \frac{\beta^\alpha u^{\alpha-1} e^{-\beta u}}{\Gamma(\alpha)}, \quad \alpha, \beta > 0, \quad (4)$$

$$\tilde{g}_{\text{wb}}(u; \alpha, \lambda) = \frac{\alpha}{\lambda} \left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-(u/\lambda)^\alpha}, \quad \alpha, \lambda > 0, \quad (5)$$

$$\tilde{g}_{\text{lmx}}(u; \alpha, c) = (\alpha - 1) c^{\alpha-1} (u + c)^{-\alpha}, \quad \alpha > 1, c > 0. \quad (6)$$

The corresponding CDFs $\tilde{F}(u; \psi) = \int_0^u \tilde{g}(v; \psi) dv$ have closed form in every case —exponential: $1 - e^{-\beta u}$; Gamma: $\gamma(\alpha, \beta u)/\Gamma(\alpha)$ (regularised lower incomplete gamma); Weibull: $1 - e^{-(u/\lambda)^\alpha}$; power-law/Lomax: $1 - (c/(u + c))^{\alpha-1}$ — and Ruhela uses this in Section 3 to evaluate the integrated intensity analytically.

For the baseline Ruhela take a log-link parametrisation

$$\nu(t; \alpha) = \exp\left(\alpha_0 + \alpha_1 \frac{t}{T} + \alpha_2 \sin \frac{2\pi t}{365.25} + \alpha_3 \cos \frac{2\pi t}{365.25}\right), \quad (7)$$

with the special case $\alpha_1 = \alpha_2 = \alpha_3 = 0$ giving the classical constant baseline $\nu(t) \equiv e^{\alpha_0}$.

3 Likelihood, score, information

For an inhomogeneous point process the joint density of the event times $\{t_i\}_{i=1}^n$ on $[0, T]$ is, up to a multiplicative constant,

$$\ell(\theta) = \sum_{i=1}^n \log \lambda(t_i; \theta) - \int_0^T \lambda(s; \theta) ds. \quad (8)$$

The integrated intensity factors as

$$\int_0^T \lambda(s; \theta) ds = \int_0^T \nu(s; \alpha) ds + \eta \sum_{i=1}^n \tilde{F}(T - t_i; \psi), \quad (9)$$

which is what makes the likelihood tractable in spite of the non-Markovian intensity: the cluster-weight contribution to $\int \lambda$ collapses to a one-dimensional sum over events. For the constant baseline $\int_0^T \nu = e^{\alpha_0} T$; for the sinusoidal baseline Ruhela uses a trapezoidal rule on a daily grid.

The score is

$$S(\theta) = \partial_\theta \ell = \sum_{i=1}^n \frac{\partial_\theta \lambda(t_i)}{\lambda(t_i)} - \int_0^T \partial_\theta \lambda(s) ds, \quad (10)$$

and the (observed) information matrix is

$$I(\theta) = -\partial_\theta^{\otimes 2} \ell = \sum_{i=1}^n \left[\frac{\partial_\theta \lambda(t_i) \partial_\theta \lambda(t_i)^\top}{\lambda(t_i)^2} - \frac{\partial_\theta^{\otimes 2} \lambda(t_i)}{\lambda(t_i)} \right] + \int_0^T \partial_\theta^{\otimes 2} \lambda(s) ds, \quad (11)$$

where $\partial_\theta = (\partial/\partial \alpha_0, \dots, \partial/\partial \eta, \partial/\partial \psi_1, \dots)$ stacks the gradients with respect to the baseline parameters α , the branching ratio η , and the kernel parameters ψ . In implementation Ruhela maximise (8) directly with adaptive Nelder-Mead and obtain $I(\hat{\theta})$ from a numerical Hessian.

4 Asymptotic theory

Following [15, 16] Ruhela adopt an *infill* asymptotic regime: a sequence $\tilde{N}^n$ of Hawkes processes on $[0, 1]$ with intensities

$$\tilde{\lambda}_t^n = n\nu(t) + \int_0^{t-} n g(n(t-s)) d\tilde{N}_s^n, \quad t \in [0, 1], \quad (12)$$

where the parameters of ν and g live in a compact set $\Theta \subset \mathbb{R}^d$. After the time change $N_t^n := \tilde{N}_{t/n}^n$ the process lives on $[0, n]$ with stationary-looking intensity $\lambda_t^n = \nu(t/n) + \int_0^{t-} g(t-s) dN_s^n$ [16, §2].

Definition 1 (*C*-ergodicity, after Clinet-Yoshida 2017). A process X is *C*-ergodic for a family C of test functions if for every $\psi \in C$ there exists $\pi(\psi) \in \mathbb{R}$ with $\frac{1}{T} \int_0^T \psi(X_s) ds \xrightarrow{\mathbb{P}} \pi(\psi)$ as $T \rightarrow \infty$.

Theorem 2 (Asymptotic ergodicity; [16], Thm. 1). *Under regularity conditions on $\nu(\cdot; \alpha)$ and $\tilde{g}(\cdot; \psi)$ (continuity, bounded moments, $\eta < 1$), the intensity process λ^n and its first two parameter derivatives $\partial_\theta \lambda^n, \partial_\theta^{\otimes 2} \lambda^n$ are asymptotically ergodic, in the sense that for every continuous polynomial-growth test function φ , $\frac{1}{n} \int_0^n \varphi(\lambda_s^n, \partial_\theta \lambda_s^n, \partial_\theta^{\otimes 2} \lambda_s^n) ds$ converges in probability to a deterministic limit.*

Theorem 3 (Consistency of MLE; [16], Thm. 2). *Under the assumptions of Theorem 2, the maximiser $\hat{\theta}^n$ of the normalised log-likelihood $\mathcal{L}^n(\theta) = \frac{1}{n} \ell^n(\theta)$ satisfies $\hat{\theta}^n \xrightarrow{\mathbb{P}} \theta_0$, where θ_0 is the unique maximiser of the deterministic limit identified by Theorem 2.*

Theorem 4 (Asymptotic normality of MLE; [16], Thm. 3). *Under the assumptions of Theorem 2 and a local non-degeneracy condition on the limit information $I(\theta_0)$,*

$$\sqrt{n}(\hat{\theta}^n - \theta_0) \xrightarrow{\mathcal{D}} \mathcal{N}(0, I(\theta_0)^{-1}). \quad (13)$$

For the finite- T application Ruhela uses (13) as a plug-in standard error: $\widehat{\text{Cov}}(\hat{\theta}) = I(\hat{\theta})^{-1}/n$ via the numerical Hessian of (8). This is exactly the inference one would have used under the classical Markovian assumption — the Kwan-Chen-Dunsmuir contribution is the proof that it remains valid when the kernel is non-exponential and the baseline is non-stationary.

5 Goodness-of-fit by time-rescaling

The Daley-Vere-Jones / Brown-Frank-Mitra time-rescaling theorem [3, 6] states that if $\Lambda(t) = \int_0^t \lambda(s) ds$ is the integrated intensity under the true (or correctly specified) model then the rescaled events $\Lambda(t_i)$ form a homogeneous unit-rate Poisson process. Equivalently, the increments $\Delta_i = \Lambda(t_i) - \Lambda(t_{i-1})$ are iid $\text{Exp}(1)$, and

$$U_i = 1 - e^{-\Delta_i} \sim \text{Uniform}(0, 1) \quad \text{iid.} \quad (14)$$

Ruhela compute the U_i for each fitted model using (9) and test (14) with the Kolmogorov-Smirnov statistic $D_n = \sup_{u \in [0, 1]} |\hat{F}_n(u) - u|$. A non-rejection ($p \geq 0.05$) is consistent with the model being correctly specified; rejections of the classical Markovian Hawkes in favour of a non-Markovian alternative are exactly the empirical claim of this paper.

6 Implementation

The new `moirais.tps_hawkes_advanced` module exposes:

- `fit_hawkes_general(t, T, kernel_kind, baseline_kind)` — MLE via adaptive Nelder-Mead with parameter-vector packing $\theta = (\alpha, \eta, \psi)$ and the integrated-intensity decomposition (9).
- `hawkes_advanced_fit(df, kernel, baseline)` — `RichResult` wrapper with Q-Q plot, KS p -value, and AIC/BIC.

- `compare_hawkes_kernels(df)` — fits all 8 (kernel $\times$ baseline) combinations and ranks by AIC.
- `hawkes_markovian_vs_nonmarkovian(df)` — focused 2-way comparison: (exp, const) vs (gamma, sin), the cheapest pairwise contrast on the dashboard.

The companion `moirais.tps_stochastic.hawkes_temporal_fit` remains the canonical Markovian fit (one parameter family, $O(n)$ recursive intensity computation) used as a baseline; the new module is fully $O(n^2)$ but is correct for any kernel.

7 Application: Toronto Police Service

Ruhela fit all eight (kernel $\times$ baseline) combinations to the TPS Assault open-data feed restricted to the post-2014 era ($n = 151\,675$ events, sub-sampled to $n = 2\,000$ for the $O(n^2)$ MLE), with $T \approx 4,138$ days (2014-01-01 to early 2026). Because OCC_DATE is daily-resolution Ruhela add $U(0,1)$ -day jitter before fitting to break ties; the original event ordering by day is preserved. Table 1 reports the AIC and Kolmogorov-Smirnov p -value for the time-rescaling residuals (14) for each combination. The Markovian classical model is the bold row.

Table 1: Kernel $\times$ baseline AIC ranking on TPS Assault (2014–2026, $n = 2\,000$ sub-sampled events, $T \approx 4,138$ days). Lower AIC is better. $\hat{\eta}$ is the fitted branching ratio.

Kernel	Baseline	k	AIC	$\hat{\eta}$	KS p
weibull	sinusoidal	7	5 352.1	0.973	0.261
exponential	sinusoidal	6	5 357.9	0.974	0.851
lomax	sinusoidal	7	5 402.6	0.839	0.841
gamma	sinusoidal	7	5 402.7	0.829	0.836
exponential	constant	3	5 493.2	0.981	0.860
weibull	constant	4	5 494.2	0.981	0.676
gamma	constant	4	5 495.2	0.981	0.859
lomax	constant	4	5 495.2	0.981	0.860

The four sinusoidal-baseline rows beat every constant-baseline row. The best fit (Weibull kernel, sinusoidal baseline) improves on the Markovian classical Hawkes by $\Delta\text{AIC} = 141.1$, a very substantial margin even after the four extra parameters of the sinusoidal baseline. By contrast, switching only the kernel (holding the baseline constant) gains essentially nothing ($\Delta\text{AIC} \leq 1.0$). The substantive conclusion is that for Toronto Assault data the dominant departure from the classical model is the time-varying baseline, not the kernel shape — but the KCD framework remains the inferential engine that makes the comparison legitimate.

Markovian vs non-Markovian. The (weibull, sinusoidal) winner encodes *both* non-Markovian generalisations simultaneously: the kernel is no longer memoryless ($\hat{\alpha}_{\text{wb}} > 1$ shifts the kernel mode away from $u = 0$) and the baseline is no longer constant ($\hat{\alpha}_2, \hat{\alpha}_3$ encode an annual-cycle modulation of $\nu(t)$). All four sinusoidal-baseline fits dominate the Markovian classical, and the within-baseline gap between kernels is small (≤ 50 AIC), so for these data the time-varying baseline is the dominant non-Markovian feature.

Branching ratio and explosivity. Across all eight fits $\hat{\eta}$ sits in $[0.83, 0.98]$ — high but stationary. That each Toronto Assault incident contributes an estimated $\hat{\eta} < 1$ direct offspring on average to the rate of further Assault incidents is consistent with the *near-repeat-victimisation* literature: criminal events near in time and space are not independent, and the self-exciting component is the formal way to score that.

8 Discussion

The contribution of this paper is methodological rather than substantive. Toronto crime data have been modelled as Hawkes processes since the original Mohler analysis of LA burglary data [4];

what has not been done, until now, is to relax both the exponential-kernel *and* the constant-baseline assumption *simultaneously* on a Toronto open-data feed. The Kwan-Chen-Dunsmuir asymptotic framework gives the inferential ground for doing so, and Ruhela’s `moirais.tps_hawkes_advanced` implementation makes it routine.

A natural next step is to lift the kernel and baseline to the spatial-marked process — already partially done in `moirais.tps_spatial_advanced` for Moran’s I / LISA / Getis-Ord but not yet for the self-exciting kernel itself. A second extension is the multivariate-marked Hawkes: cross-excitation between Assault, Robbery, and Auto-Theft can be encoded as a $K \times K$ kernel matrix $g_{kl}(\cdot)$, with the same Kwan-Chen-Dunsmuir conditions applying coordinate-wise.

9 Conclusion

Ruhela have given a self-contained derivation of the likelihood, score, and information for a non-stationary, non-Markovian Hawkes process, summarised the asymptotic guarantees of Kwan-Chen-Dunsmuir, and implemented the full machinery in MOIRAIS. Markovian and non-Markovian fits are now comparable on the same footing, both on synthetic stress-tests and on Toronto Police Service incident feeds. The classical Mohler model is no longer the only Hawkes model in the toolbox.

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